Adding and Subtracting in Scientific Notation

Teacher's Edition

The Period

MA.8.NSO.1.5 Add, subtract, multiply and divide numbers expressed in scientific notation with procedural fluency.

Must Add and subtract numbers written in scientific notation with procedural fluency, and write every answer with a coefficient at least 1 and less than 10.

Must not The two exponents stay within 2 of each other — that is this benchmark's own limit. Multiplication and division are 4.02–03; significant digits are 4.04.

Target I can add and subtract two numbers written in scientific notation, and write the answer in scientific notation.

MTR **MTR.5.1** — Notes I · **MTR.6.1** — Boards 3 and 4 · **MTR.4.1** — Board 6

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	11 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	3 min
Your Turn	14–15	3 min
Whiteboards	16–33	12 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. The powers of ten have to match before a single digit is added. That is the whole lesson.
2. The power of ten is a label, like the word thousands. You never add the label.
3. When you finish, look at the coefficient. If it is not between 1 and 10, you are not done.

Slide by Slide

1. **Adding and Subtracting in Scientific Notation.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is today's lesson in miniature. *1 retrieves 3.09 — small numbers in scientific notation: the 6 is in the ten-thousandths place (yesterday). 2 retrieves 3.08 — divide the coefficients, divide the powers, multiply the results (last week). 3 retrieves Unit 2 — perfect squares (2.01). Roots come back for good in 4.05 (last unit). 4 retrieves MA.5.NSO.2.3 — adding decimals: line up the place values, not the digits (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 6.2×10^{-4} 2. 4,000 3. 12 4. 1.69
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the table.** You can only add things that are the same size. Ten-thousands add to ten-thousands, and the label never changes.
6. **Notes.** The power of ten is a label. Six thousands plus two thousands is eight thousands — you never add the word thousands.
7. **Notes.** The exponents are never added. Adding 10 to the fifth and 10 to the third does not make 10 to the eighth — it does not make a power of ten at all.
8. **Example 1 — Same power, and the answer is not finished.** SPOKEN ONLY.
9. **Example 1 — Worked.** Add the coefficients, keep 10^7, then repair.
10. **Your Turn — Same steps, your numbers. Boards up when done.** Same shape with negative exponents.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Different powers, and a subtraction.** SPOKEN ONLY.
13. **Example 2 — Worked.** Rewrite 4.5×10^5 over 10^6 — exponent up one, point one place left — and the subtraction becomes ordinary.
14. **Your Turn — Same steps, your numbers. Boards up when done.** The exponents are two apart — the widest gap this benchmark allows.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. **Board 1.** $(3.2\times10^5)+(4.1\times10^5)$ → ***7.3 × 10⁵*** *7.3×10^{10} — the exponents were added as well as the coefficients [B1G-M 8.NSO.1.5 misconception: exponents added]; 1.312×10^{11} — the numbers were multiplied [8.NSO.1.5: the operation is addition]*

18–19. **Board 2.** A stadium sold 4.2×10^4 tickets on Friday and 3.5×10^4 on Saturday. Each ticket cost 38 dollars. How many tickets were sold on the two days? → ***7.7 × 10⁴*** *The ticket price of 38 dollars is not needed. anything built from 38 — the unneeded figure was used [the question asks for a count, not a cost]; 7.7×10^8 — the exponents were added [B1G-M 8.NSO.1.5 misconception]*

20–21. **Board 3.** $(8\times10^{-6})+(3\times10^{-6})$ → ***1.1 × 10⁻⁵*** *11×10^{-6} — right value, but 11 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving 12×10^9 instead of 1.2×10^{10}]; 1.1×10^{-11} — the exponents were added, giving 11×10^{-12}, and the answer was then finished*

22–23. **Board 4.** $(5.6\times10^8)-(4.9\times10^8)$ → ***7 × 10⁷*** *0.7×10^8 — right value, but 0.7 is not at least 1 [8.NSO.1.5: the coefficient satisfies $1 \le a < 10$]; 1.05×10^9 — added instead of subtracting*

24–25. Board 5. $(1.6 \times 10^4) + (2.3 \times 10^6)$. Which is the sum? → ***C — 2.316 × 10⁶***
(A) the coefficients were added without matching the powers of ten first [8.NSO.1.5: the powers must match]; (B) the coefficients were added and the exponents were added [B1G-M 8.NSO.1.5 named misconception: $(1.3 \times 10^3) + (3.4 \times 10^5) = 4.7 \times 10^8$]; (D) 1.6×10^4 was rewritten as 0.16×10^6 — the decimal point moved one place too few [8.NSO.1.5: the point moves as many places as the exponent changes] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

26–27. Board 6. $(2.5 \times 10^6) + (9.8 \times 10^6)$. Which is the sum, written in scientific notation? → ***D — 1.23 × 10⁷*** *(A) equal in value, but 12.3 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving 12×10^9 instead of 1.2×10^{10}]; (B) the exponents were added as well as the coefficients [8.NSO.1.5: the power of ten does not change]; (C) the coefficient was repaired but the exponent was not raised [8.NSO.1.5: point one place left, exponent up one] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. $(7.8\times10^{-2})-(6.4\times10^{-3})$ → ***7.16 × 10⁻²*** *1.4×10^{-2} — the coefficients were subtracted without matching the powers [8.NSO.1.5]; 7.16×10^{-5} — the exponents were added, $-2 + (-3)$ [8.NSO.1.5: the power of ten is a label, not a term]*

30–31. Board 8. $(4.5 \times 10^6) + (n \times 10^6) = 1.02 \times 10^7$. What is the value of n? → ***n = 5.7*** *−3.48 — the target was not rewritten over 10^6 before subtracting [8.NSO.1.5: match the powers first]; 5.7×10^6 — the blank holds a coefficient, not a whole term*

32–33. Board 9. Devon writes $(1.9 \times 10^2) + (6.6 \times 10^4) = 8.5 \times 10^6$. Name the two mistakes he made, and give the correct sum. → ***6.619 × 10⁴*** *'he added wrong' — no mistake is actually named [the item asks which two]; 8.5×10^4 — the exponent mistake was caught and the matching mistake was not*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Misconceptions to Watch

Board 1. 7.3×10^{10} — the exponents were added as well as the coefficients [B1G-M 8.NSO.1.5 misconception: exponents added]

Board 2. anything built from 38 — the unneeded figure was used [the question asks for a count, not a cost]

Board 3. 11×10^{-6} — right value, but 11 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving 12×10^9 instead of 1.2×10^{10}]

Board 4. 0.7×10^8 — right value, but 0.7 is not at least 1 [8.NSO.1.5: the coefficient satisfies $1 \le a < 10$]

Board 5. the coefficients were added without matching the powers of ten first [8.NSO.1.5: the powers must match]

Board 6. equal in value, but 12.3 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving 12×10^9 instead of 1.2×10^{10}]

Board 7. 1.4×10^{-2} — the coefficients were subtracted without matching the powers [8.NSO.1.5]

Board 8. −3.48 — the target was not rewritten over 10^6 before subtracting [8.NSO.1.5: match the powers first]

Board 9. 'he added wrong' — no mistake is actually named [the item asks which two]

Materials Whiteboards and markers. Calculators are allowed on everything; entering 8E−6 + 3E−6 and reading 1.1E−5 back off the display is worth thirty seconds at whiteboard 3.

Multiplying and Dividing in Scientific Notation

Teacher's Edition

The Period

MA.8.NSO.1.5 Add, subtract, multiply and divide numbers expressed in scientific notation with procedural fluency.

Must Multiply and divide numbers written in scientific notation with procedural fluency, and write every answer with a coefficient at least 1 and less than 10.

Must not No within-2 limit applies here — that clarification is about addition and subtraction only, and the contrast with 4.01 is the point of the day. Significant digits are 4.04.

Target I can multiply and divide two numbers written in scientific notation, and write the answer in scientific notation.

MTR **MTR.5.1** — Notes I · **MTR.3.1** — Boards 1, 3 and 7 · **MTR.4.1** — Board 6

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	11 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	3 min
Your Turn	14–15	3 min
Whiteboards	16–33	12 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. The powers of ten never have to match when you multiply or divide. They can be as far apart as they like.
2. Multiply the fronts and add the backs; divide the fronts and subtract the backs, top minus bottom.
3. A product of two coefficients under 10 is very often over 10. Look at the coefficient before you put the board up.

Slide by Slide

1. **Multiplying and Dividing in Scientific Notation.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q1 and Q2 are the two halves of today, side by side. *1 retrieves 4.01 — same power: add the coefficients and keep the power (yesterday). 2 retrieves 3.02 — quotient of powers with base 10: subtract the exponents (last week). 3 retrieves Unit 2 — perfect squares (2.01). Roots return in 4.05 (last unit). 4 retrieves MA.6.NSO.2.1 — dividing decimals; today every quotient starts with one of these (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 6.2×10^6 2. 10^5 3. 13 4. 3
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy both lines.** Multiply the fronts, add the backs. The powers of ten never have to match when you multiply.
6. **Notes.** Top minus bottom, always in that order. And subtracting a negative goes up.
7. **Notes.** Coefficient over 10, point left and exponent up. Coefficient under 1, point right and exponent down. Same repair as yesterday.
8. **Example 1 — Multiplying, and the answer needs finishing.** SPOKEN ONLY.
9. **Example 1 — Worked.** Regroup, multiply, add the exponents, then repair.
10. **Your Turn — Same steps, your numbers. Boards up when done.** Same shape, the other sign on the small exponent.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Multiplication and division in one expression.** SPOKEN ONLY.
13. **Example 2 — Worked.** The intermediate value is 60×10^6, which is exactly the unfinished form — and it does not have to be repaired yet, because the division is still to come.
14. **Your Turn — Same steps, your numbers. Boards up when done.** A negative exponent inside the product and a division that pushes the answer below 1.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
16. **–17. Board 1.** $(3\times10^5)(3\times10^6)$ → ***9 × 10¹¹*** *9×10^{30} — the exponents were multiplied [B1G-M 8.NSO.1.5 named misconception: $(2 \times 10^4)(3 \times 10^5) = 6 \times 10^{20}$]; 6×10^{11} — the coefficients were added [8.NSO.1.5: the operation is multiplication]*
18. **–19. Board 2.** A printer makes 4×10^3 copies per hour. It runs for 2.5×10^2 hours. Paper costs 3 cents a sheet. How many copies does it make? → ***1 × 10⁶*** *The paper cost of 3 cents a sheet is not needed. anything built from 3 cents — the unneeded figure was used [the question asks for a count, not a cost]; 10×10^5 — right value, coefficient not less than 10 [B1G-M 8.NSO.1.5 misconception]*
20. **–21. Board 3.** $\frac{9.6\times10^8}{1.2\times10^3}$ → ***8 × 10⁵*** *8×10^{11} — the exponents were added [8.NSO.1.5: division subtracts the exponents]; 8.4×10^5 — the coefficients were subtracted instead of divided [8.NSO.1.5]*

22–23. Board 4. $\frac{3.28\times10^{7}}{4\times10^{10}}$ → ***8.2 × 10⁻⁴*** *0.82 × 10⁻³ — right value, but 0.82 is not at least 1 [8.NSO.1.5: the coefficient satisfies 1 ≤ a < 10]; 8.2 × 10⁻³ — the point moved but the exponent did not*

24–25. Board 5. (4 × 10⁷)(2 × 10⁹). Which is the product? → ***C — 8 × 10¹⁶*** *(A) the exponents were subtracted, 7 − 9 [8.NSO.1.5: multiplication adds the exponents]; (B) the exponents were multiplied [B1G-M 8.NSO.1.5 named misconception: (2 × 10⁴)(3 × 10⁵) = 6 × 10²⁰]; (D) the coefficients were added instead of multiplied [8.NSO.1.5: the operation is multiplication] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

26–27. Board 6. (5 × 10⁶)(4 × 10³). Which is the product, written in scientific notation? → ***D — 2 × 10¹⁰*** *(A) the coefficient was repaired but the exponent was not raised [8.NSO.1.5: point one place left, exponent up one]; (B) equal in value, but 20 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving 12 × 10⁹ instead of 1.2 × 10¹⁰]; (C) the exponents were multiplied and the coefficient was left over 10 [8.NSO.1.5] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. $\frac{7\times10^{9}}{7\times10^{-6}}$ → ***1 × 10¹⁵*** *1 × 10³ — the negative sign on the bottom exponent was dropped [8.NSO.1.5: subtracting a negative adds]; 1 × 10⁻¹⁵ — bottom minus top [8.NSO.1.5: top minus bottom]*

30–31. Board 8. (8.4 × 10⁷) ÷ (n × 10²) = 4.2 × 10⁵. What is the value of n? → ***n = 2*** *4.2 — the coefficients were subtracted, 8.4 − 4.2 [8.NSO.1.5: the coefficients divide]; 0.5 — 4.2 ÷ 8.4 instead of 8.4 ÷ 4.2*

32–33. Board 9. For (7.2 × 10¹¹) ÷ (2.4 × 10³), Marcus writes 4.8 × 10⁸ and Sara writes 3 × 10¹⁴. Name what each one did wrong, and give the correct quotient. → ***3 × 10⁸*** *'they both got it wrong' — neither error is named [the item asks what each one did]; 4.8 × 10⁸ — Marcus's error repeated*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Misconceptions to Watch

Board 1. 9 × 10³⁰ — the exponents were multiplied [B1G-M 8.NSO.1.5 named misconception: (2 × 10⁴)(3 × 10⁵) = 6 × 10²⁰]

Board 2. anything built from 3 cents — the unneeded figure was used [the question asks for a count, not a cost]

Board 3. 8 × 10¹¹ — the exponents were added [8.NSO.1.5: division subtracts the exponents]

Board 4. 0.82 × 10⁻³ — right value, but 0.82 is not at least 1 [8.NSO.1.5: the coefficient satisfies 1 ≤ a < 10]

Board 5. the exponents were subtracted, 7 − 9 [8.NSO.1.5: multiplication adds the exponents]

Board 6. the coefficient was repaired but the exponent was not raised [8.NSO.1.5: point one place left, exponent up one]

Board 7. 1 × 10³ — the negative sign on the bottom exponent was dropped [8.NSO.1.5: subtracting a negative adds]

Board 8. 4.2 — the coefficients were subtracted, 8.4 − 4.2 [8.NSO.1.5: the coefficients divide]

Board 9. 'they both got it wrong' — neither error is named [the item asks what each one did]

Materials Whiteboards and markers. Calculators are allowed on everything; a calculator is the fastest way to settle whether 8×10^{16} and 8×10^{63} could possibly both be the answer to whiteboard 5.

Real-World Problems and Significant Digits

Teacher's Edition

The Period

MA.8.NSO.1.6 Solve real-world problems involving operations with numbers expressed in scientific notation.

Must Solve real-world problems with numbers in scientific notation, and recognise the importance of significant digits when physical measurements are involved.

Must not The within-2 exponent limit still applies to addition and subtraction here. Significant digits limit a reported answer; they never change the arithmetic, and a count is never a measurement.

Target I can solve a real-world problem with numbers in scientific notation and report the answer with the right number of significant digits.

MTR **MTR.6.1** — Boards 2, 6 and 7 · **MTR.7.1** — Every problem from Notes II on · **MTR.4.1** — Board 6

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	12 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	3 min
Your Turn	14–15	3 min
Whiteboards	16–33	11 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. A measurement's last digit is uncertain. That is what the whole rule is protecting.
2. Products and quotients count digits. Sums and differences count places. Two rules, and it matters which one you reach for.
3. Do all the arithmetic first and round once, at the end. Rounding in the middle loses digits the arithmetic still needed.

Slide by Slide

1. **Real-World Problems and Significant Digits.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q3 and Q4 are today's two tools. *1 retrieves 4.02–03 — multiply the coefficients, add the exponents, then finish the answer (yesterday). 2 retrieves 4.01 — rewrite one number over the other's power, then subtract the coefficients (last week). 3 retrieves 3.09 — small numbers; note that the zero between the 9 and the 2 stays (last unit). 4 retrieves MA.5.NSO.1.5 — rounding to a named place; every answer today ends with one of these (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 1.2×10^{4} 2. 5.89×10^{7} 3. 9.02×10^{-3} 4. 4.74
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the table.** Scientific notation makes this easy. In 5.2 times ten to the fourth, every digit you can see is significant — that is one of the reasons the notation exists.
6. **Notes.** Two significant digits in, two significant digits out. The answer is never better than the measurement it came from.
7. **Notes — Copy all three lines.** Products count digits. Sums count places. They are two different rules and it matters which one you reach for.
8. **Example 1 — A product of two measurements.** SPOKEN ONLY.
9. **Example 1 — Worked.** Multiply in full — 10,750 — and round once.
10. **Your Turn — Same steps, your numbers. Boards up when done.** Same shape, the counts the other way round.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — A difference of two measurements.** SPOKEN ONLY.
13. **Example 2 — Worked.** Subtract in full, then stop at the leftmost last-digit place.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 3,850 stops at the tens place and 274 stops at the ones, so the answer stops at the tens.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. How many significant digits are in 0.00470 ? → ***3*** *2 — the trailing zero was dropped [B1G-M 8.NSO.1.6: a trailing zero after a decimal point is significant]; 5 — the leading zeros were counted [B1G-M 8.NSO.1.6: leading zeros are never significant]*

18–19. Board 2. A pond's surface measures 1.4×10^{2} m by 8.5×10^{1} m. The pond is 3 meters deep. Find the surface area. Round to the correct number of significant digits. → ***1.2×10^{4} square meters*** *The depth of 3 meters is not needed. anything built from the depth — the unneeded figure was used [an area needs two dimensions]; 1.19×10^{4} — the arithmetic without the rounding [8.NSO.1.6 clarification]*

20–21. Board 3. Two measurements are multiplied. One has 3 significant digits and the other has 5. The exact product is 7.4126×10^{5}. How should the answer be written? → ***7.41×10^{5}*** *7.4×10^{5} — two digits, the wrong one of the two counts [8.NSO.1.6: the answer takes the smaller count, which is 3]; 7.4126×10^{5} — no rounding at all*

22–23. Board 4. $(2.5 \times 10^3) + (4.71 \times 10^2)$ meters, both measured. Write the sum with the correct precision. → ***3.0 × 10³ m*** *2.971 × 10³ — the arithmetic without the precision rule [8.NSO.1.6 clarification]; 2.9 × 10³ — 2,971 rounded down instead of up [rounding to the hundreds place]*

24–25. Board 5. How many significant digits are in 0.03080? → ***A — 4*** *(B) only the nonzero digits were counted [B1G-M 8.NSO.1.6: a zero between nonzero digits is always significant]; (C) one of the leading zeros was counted [B1G-M 8.NSO.1.6: leading zeros are never significant]; (D) the trailing zero was dropped [B1G-M 8.NSO.1.6: a trailing zero after a decimal point is significant] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

26–27. Board 6. A rectangle measures 6.0×10^2 cm by 4.35×10^2 cm. Which is the area, to the correct significant digits? → ***B — 2.6 × 10⁵ cm²*** *(A) the exact product, not reduced, and 26.1 is not less than 10 — two mistakes [B1G-M 8.NSO.1.5 misconception: leaving 12×10^9 instead of 1.2×10^{10}]; (C) the exponent was not raised when the coefficient was repaired [8.NSO.1.5: point one place left, exponent up one]; (D) the exact product, not reduced to the two significant digits the measurements carry [8.NSO.1.6 clarification] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. A car travels 4.20×10^2 km in 3.0 hours. What is its average speed, to the right number of significant digits? → ***1.4 × 10² km/h*** *140.0 — four digits claimed from a two-digit measurement [8.NSO.1.6 clarification]; 1.4×10^1 — the exponent is one too low*

30–31. Board 8. A measured length times a measured width gives an area of 3.6×10^3 m². The length measured 9.0×10^1 m. What did the width measure, to two significant digits? → ***4.0 × 10¹ m*** *4×10^1 — one significant digit where two were asked for [8.NSO.1.6: a trailing zero after a decimal point carries the second digit]; 4.0×10^2 — the exponent is one too high*

32–33. Board 9. Ellis says that because 9,145 has 4 significant digits and 4.8 has 2, the sum 9,145 + 4.8 must be written with 2 significant digits, as 9.1×10^3. Explain why he is wrong, and give the correct sum. → ***9,150 = 9.150 × 10³*** *'he used the wrong rule' — the right rule is not stated [the item asks for an explanation]; 9,149.8 — no rounding at all, which claims a tenth that 9,145 never measured*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Misconceptions to Watch

Board 1. 2 — the trailing zero was dropped [B1G-M 8.NSO.1.6: a trailing zero after a decimal point is significant]

Board 2. anything built from the depth — the unneeded figure was used [an area needs two dimensions]

Board 3. 7.4×10^5 — two digits, the wrong one of the two counts [8.NSO.1.6: the answer takes the smaller count, which is 3]

Board 4. 2.971×10^3 — the arithmetic without the precision rule [8.NSO.1.6 clarification]

Board 5. only the nonzero digits were counted [B1G-M 8.NSO.1.6: a zero between nonzero digits is always significant]

Board 6. the exact product, not reduced, and 26.1 is not less than 10 — two mistakes [B1G-M 8.NSO.1.5 misconception: leaving 12×10^9 instead of 1.2×10^{10}]

Board 7. 140.0 — four digits claimed from a two-digit measurement [8.NSO.1.6 clarification]

Board 8. 4×10^{1} — one significant digit where two were asked for [8.NSO.1.6: a trailing zero after a decimal point carries the second digit]

Board 9. 'he used the wrong rule' — the right rule is not stated [the item asks for an explanation]

Materials Whiteboards and markers. Calculators are allowed on everything and are genuinely useful today, because the point of the lesson is what to do with a long display rather than how to produce one.

Evaluating Expressions with Radicals

Teacher's Edition

The Period

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

Must Evaluate expressions containing square roots, cube roots and integer exponents, simplifying radicals by factoring within the benchmark's lists.

Must not Perfect squares up to 225 and perfect cubes from −125 to 125, or fractions whose parts are each in those lists. Estimating a root that is not perfect was Unit 2. No fractional exponents.

Target I can evaluate an expression that contains square roots, cube roots and integer exponents.

MTR **MTR.5.1** — Notes I · **MTR.6.1** — Boards 3 and 6 · **MTR.4.1** — Board 5

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	12 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	3 min
Your Turn	14–15	3 min
Whiteboards	16–33	11 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. The radical bar is a pair of parentheses you cannot see. Finish what is under it, then take the root.
2. A cube root of a negative number is ordinary. A square root of a negative number is not a number you have.
3. A negative exponent is still a reciprocal. Nothing about the radical changes that.

Slide by Slide

1. **Evaluating Expressions with Radicals.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q3 and Q4 are the two halves of today. *1 retrieves 4.04 — leading zeros never count; the trailing zero does, because a decimal point is written (yesterday). 2 retrieves 4.02–03 — multiply the coefficients, add the exponents (last week). 3 retrieves Unit 2 — perfect squares (2.01): 14 · 14 = 196 (last unit). 4 retrieves MA.5.AR.2.2 — multiply before you subtract; today the radical joins that list (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 3 2. 6×10^{-3} 3. 14 4. 8
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy all three lines.** Whatever sits under the bar is inside a pair of parentheses you cannot see. Finish it, then take the root.
6. **Notes.** Cube roots of negative numbers are ordinary. Square roots of negative numbers are not numbers you have — that is the difference to hold on to.
7. **Notes.** A negative exponent is still a reciprocal. The radical does not change a single thing about that.
8. **Example 1 — A power and a radical in one expression.** SPOKEN ONLY.
9. **Example 1 — Worked.** Two independent pieces.
10. **Your Turn — Same steps, your numbers. Boards up when done.** The guide's tiered version: a cube root this time, and a subtraction.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — A radical inside a product.** SPOKEN ONLY.
13. **Example 2 — Worked.** Three layers, innermost first: under the bar, then the root, then the multiplication, then the subtraction.
14. **Your Turn — Same steps, your numbers. Boards up when done.** A negative cube root and a product that grows.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. $\sqrt{121}$ → ***11*** *60.5 — the radicand was halved [Unit 2, 2.02: a root is not a half]; ±11 — the radical sign names the principal root only [8.NSO.1.7]*

18–19. Board 2. A square patio has an area of 169 square feet. Fencing for it costs 12 dollars a foot. How long is one side of the patio? → ***13 feet*** *The fencing price of 12 dollars a foot is not needed. anything built from 12 dollars — the unneeded figure was used [the question asks for a length, not a cost]; 52 — the perimeter, not a side; 84.5 — the area was halved*

20–21. Board 3. $\sqrt[3]{-125}$ → ***−5*** *5 — the sign was dropped [8.NSO.1.7: an odd root keeps the sign of its radicand]; 'not a real number' — the square-root rule applied to a cube root [B1G-M 8.NSO.1.7 misconception: confusing square roots with cube roots]*

22–23. Board 4. $\sqrt{2^3 + 17}$ → ***5*** *25 — the arithmetic under the bar was done and the root was not [8.NSO.1.7]; anything with two separate roots — the bar groups, so it is one radicand [8.NSO.1.7]*

24–25. Board 5. √(25 + 144). Which is the value? → ***D — 13*** *(A) the root of each piece was taken and the results added, 5 + 12 [8.NSO.1.7: the bar groups, so the radicand is 169]; (B) the sum was found and the root was never taken [8.NSO.1.7]; (C) the radicand was halved instead of rooted [Unit 2, 2.02: a root is not a half] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

26–27. Board 6. The cube root of −1/64. Which is the value? → ***A — −1/4*** *(B) the square-root rule for negatives was applied to a cube root [B1G-M 8.NSO.1.7 misconception]; (C) the square root of 1/64 was taken and the sign kept [B1G-M 8.NSO.1.7 misconception: confusing square roots with cube roots]; (D) the sign was dropped [8.NSO.1.7: an odd root keeps the sign of its radicand] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. $\sqrt{\frac{4}{9}} + \sqrt[3]{\frac{8}{27}}$ → ***4/3*** *12/36 — the roots were never taken, then 4/9 + 8/27 was added top-and-bottom [two errors: the roots, then Grade 5 fraction addition]; 2/3 — only one of the two radicals was evaluated*

30–31. Board 8. $\sqrt{5^3 - 25} \cdot 10^{-3}$ → ***1/100*** *10,000 — 10^{-3} read as 10^3 [8.NSO.1.3: a negative exponent is a reciprocal]; −1/100 — the exponent's sign made the value's sign [B1G-M 8.NSO.1.3 task: −b vs b^{-1}]*

32–33. Board 9. Simon says that the cube root of −27 is not a real number, because you cannot take the root of a negative. Explain why he is wrong, and give the value. → ***−3*** *'−3' alone — the value without the reason [the item asks why he is wrong]; '3' — the sign dropped as well as the rule misread*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Misconceptions to Watch

Board 1. 60.5 — the radicand was halved [Unit 2, 2.02: a root is not a half]

Board 2. anything built from 12 dollars — the unneeded figure was used [the question asks for a length, not a cost]

Board 3. 5 — the sign was dropped [8.NSO.1.7: an odd root keeps the sign of its radicand]

Board 4. 25 — the arithmetic under the bar was done and the root was not [8.NSO.1.7]

Board 5. the root of each piece was taken and the results added, 5 + 12 [8.NSO.1.7: the bar groups, so the radicand is 169]

Board 6. the square-root rule for negatives was applied to a cube root [B1G-M 8.NSO.1.7 misconception]

Board 7. 12/36 — the roots were never taken, then 4/9 + 8/27 was added top-and-bottom [two errors: the roots, then Grade 5 fraction addition]

Board 8. 10,000 — 10^{-3} read as 10^3 [8.NSO.1.3: a negative exponent is a reciprocal]

Board 9. '−3' alone — the value without the reason [the item asks why he is wrong]

Materials Whiteboards and markers, and Unit 2's reference sheet if students still want the perfect-square list. Calculators are allowed on everything; a calculator will refuse the cube root of a negative on some models, which is worth showing once and explaining rather than trusting.

Order of Operations with Radicals

Teacher's Edition

The Period

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

Must Evaluate multi-step expressions with rational numbers, exponents and radicals, in six steps or fewer.

Must not No mnemonic. The state guide says to avoid PEMDAS because it leaves out grouping symbols, and the word appears on no student page in this unit.

Target I can evaluate a multi-step expression with rational numbers, exponents and radicals by reading its structure.

MTR **MTR.5.1** — Notes I · **MTR.3.1** — Boards 1, 3 and 7 · **MTR.4.1** — Board 5

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	12 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	3 min
Your Turn	14–15	3 min
Whiteboards	16–33	11 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. There is nothing to memorise. Read the expression, ask what is holding what, and the order tells you itself.
2. Nothing says multiply before divide. They are one level, taken left to right like a sentence.
3. A fraction bar is a grouping symbol on the top and another one on the bottom.

Slide by Slide

1. **Order of Operations with Radicals.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is the prerequisite and it is the one half the room gets wrong. *1 retrieves 4.05 — finish under the bar first: 25 − 9 = 16 (yesterday). 2 retrieves 4.04 — the middle zero counts and the trailing zero counts, because a decimal point is written (last week). 3 retrieves 3.05 — a negative exponent flips the base (last unit). 4 retrieves MA.5.AR.2.2 — multiplication and division are one level, taken left to right (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 4 2. 4 3. 9/4 4. 8
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the table.** There is nothing to memorise here. Read the expression, ask what is holding what, and the order tells you itself.
6. **Notes.** Nothing in mathematics says multiply before divide. They are the same level, and you read left to right like a sentence.
7. **Notes.** A fraction bar is a grouping symbol on the top and another one on the bottom. Finish each, then divide.
8. **Example 1 — A grouping symbol inside a grouping symbol.** SPOKEN ONLY.
9. **Example 1 — Worked.** The power is inside the parentheses, so it goes first even though it is a level below grouping — because it is inside.
10. **Your Turn — Same steps, your numbers. Boards up when done.** Same shape, a square root and a positive answer.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — A fraction bar and a radical in one expression.** SPOKEN ONLY.
13. **Example 2 — Worked.** Three groups, each finished on its own before anything crosses between them.
14. **Your Turn — Same steps, your numbers. Boards up when done.** Same three groups, and both halves land on a half.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
16. **16–17. Board 1.** $90 \div 5 \cdot 3$ → ***54*** *6 — 5 · 3 was done first [8.NSO.1.7: multiplication and division are one level, taken left to right]; 1,350 — the division was read as a multiplication*
17. **18–19. Board 2.** A square tile has side √196 centimeters. Its diagonal is about 19.8 centimeters. What is the perimeter of the tile? → ***56 centimeters*** *The diagonal of about 19.8 centimeters is not needed. anything built from 19.8 — the unneeded figure was used [a perimeter needs the side]; 14 — the side, not the perimeter; 196 — the root was never taken*
18. **20–21. Board 3.** $16 - 5 + 4$ → ***15*** *7 — 5 + 4 was done first [8.NSO.1.7: addition and subtraction are one level, taken left to right]*
19. **22–23. Board 4.** $\frac{3^3 - 21}{5^2 - 15}$ → ***3/5*** *anything that divides before the top and bottom are finished [8.NSO.1.7: the fraction bar groups both]; −6/5 — 5^2 read as 5 · 2, leaving 6 over −5; 6/25 — the −15 under the bar was dropped*

24–25. Board 5. $2 + 3 \cdot 4^2$. Which is the value? → ***B — 50*** *(A) the exponent was applied to 3 · 4 instead of to the 4 alone [8.NSO.1.7: an exponent belongs to its own base]; (C) 4^2 was read as 4 · 2 [8.NSO.1.3: an exponent is not a factor]; (D) 2 + 3 was done first, then 5 · 16 [8.NSO.1.7: multiplication comes before addition]* Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

26–27. Board 6. $(6^2 - 4)$ over 2^3. Which is the value? → ***A — 4*** *(B) only the 4 was divided by 2^3 before the subtraction [8.NSO.1.7: the bar groups the whole numerator]; (C) 6^2 was divided by 2^3 first and then 4 was subtracted [8.NSO.1.7: the bar groups the whole numerator]; (D) the numerator was multiplied by 2^3 instead of divided [8.NSO.1.7: the bar is division]* Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

28–29. Board 7. $-2^2 + (-2)^2$ → ***0*** *8 — both terms read as $(-2)^2$ [8.NSO.1.7: an exponent attaches to its own base, and parentheses decide what that base is]; −8 — both terms read as -2^2*

30–31. Board 8. $12\left(\left(\frac{1}{2}\right)^3 + 3^{-2}\right)$ → ***17/6*** *12/8 + 12/9 left unadded — a value, not an answer; 1/8 + 1/9 = 2/17 — the denominators were added [Grade 5: adding fractions needs a common denominator]*

32–33. Board 9. Roscoe says that $(\sqrt{n} - 11)^2$ is always positive, because a square is always positive. Is he right? Give a value of n that settles it. → ***No — at n = 121 the value is 0*** *'yes, squares are positive' — the claim repeated rather than tested [8.NSO.1.7: a square is never negative, which is a different statement]; 'no' with no value of n — the counterexample is the evidence*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Misconceptions to Watch

Board 1. 6 — 5 · 3 was done first [8.NSO.1.7: multiplication and division are one level, taken left to right]

Board 2. anything built from 19.8 — the unneeded figure was used [a perimeter needs the side]

Board 3. 7 — 5 + 4 was done first [8.NSO.1.7: addition and subtraction are one level, taken left to right]

Board 4. anything that divides before the top and bottom are finished [8.NSO.1.7: the fraction bar groups both]

Board 5. the exponent was applied to 3 · 4 instead of to the 4 alone [8.NSO.1.7: an exponent belongs to its own base]

Board 6. only the 4 was divided by 2^3 before the subtraction [8.NSO.1.7: the bar groups the whole numerator]

Board 7. 8 — both terms read as $(-2)^2$ [8.NSO.1.7: an exponent attaches to its own base, and parentheses decide what that base is]

Board 8. 12/8 + 12/9 left unadded — a value, not an answer

Board 9. 'yes, squares are positive' — the claim repeated rather than tested [8.NSO.1.7: a square is never negative, which is a different statement]

Materials Whiteboards and markers. Calculators are allowed on everything, and this is the lesson where a calculator is worth using against the class: enter -2^2 and $(-2)^2$ and let the machine settle it.

Real-World Order of Operations

Teacher's Edition

The Period

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

Must Solve multi-step real-world problems with the order of operations, exponents and radicals, in six steps or fewer, and report the answer with its unit.

Must not No keyword strategy and no CUBES — the state guide names circling numbers and boxing key words as a misconception, not a method. No PEMDAS.

Target I can substitute values into a formula and evaluate it, reading what the situation does rather than hunting for key words.

MTR **MTR.7.1** — Notes I–III, both Examples and boards 2, 3, 4, 6, 7 and 9 · **MTR.2.1** — Notes II and board 9 · **MTR.4.1** — Board 6

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	12 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	3 min
Your Turn	14–15	3 min
Whiteboards	16–33	11 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. Every substituted value goes into parentheses. That one habit prevents half of what goes wrong today.
2. There is no word that tells you which operation to use. Read what is happening, not which words are there.
3. Round once, at the very end, and write the unit. An answer with no unit is not an answer.

Slide by Slide

1. **Real-World Order of Operations.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is the prerequisite and it carries the day's one habit. *1 retrieves 4.06 — the exponent attaches to its own base; the parentheses decide what that base is (yesterday). 2 retrieves 4.05 — finish each piece, then combine: 12 + 1/2 (last week). 3 retrieves 3.05 — a negative exponent flips the base (last unit). 4 retrieves MA.6.AR.1.3 — substitute, then follow the structure: $2(3)^2 = 2(9)$ (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 0 2. 25/2 3. 4/3 4. 18
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy both lines.** Every substituted value goes into parentheses. That single habit prevents half of what goes wrong in this lesson.
6. **Notes — Copy both problems.** There is no word that tells you which operation to use. Read what is happening, not which words are present.
7. **Notes.** Round once, at the very end. Rounding in the middle moves the answer, and it moves it further every step you take afterwards.
8. **Example 1 — A formula with a power in it.** SPOKEN ONLY.
9. **Example 1 — Worked.** Substitute in parentheses, square the speed, then multiply.
10. **Your Turn — Same steps, your numbers. Boards up when done.** Same formula, smaller numbers.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — A cost with a rate and a flat fee.** SPOKEN ONLY.
13. **Example 2 — Worked.** The two per-square-foot prices go inside the group that is multiplied by the area; the flat fee stays outside it.
14. **Your Turn — Same steps, your numbers. Boards up when done.** One rate and one flat fee this time.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
16. **Board 1.** Evaluate $3x^2$ when $x = -2$. → ***12*** *−12 — the exponent was applied to the 2 and not to the whole substituted value [8.NSO.1.7: substitute in parentheses]; 36 — the exponent was applied to 3x [8.NSO.1.7: the exponent belongs to its own base]*
17. **Board 1.**
18. **Board 2.** A cyclist has a mass of 70 kilograms and rides at 8 meters per second. Her bicycle cost 480 dollars. Kinetic energy is $K = \frac{1}{2}mv^2$, for a mass m and a speed v. Find her kinetic energy. → ***2,240 joules*** *The price of the bicycle is not needed. anything built from 480 — the unneeded figure was used [the formula takes a mass and a speed]; 560 — v^2 read as 2v [8.NSO.1.3: an exponent is not a factor]; 280 — the square dropped altogether; 2,240 with no unit — the answer is in joules*
19. **Board 2.**
20. **Board 3.** Renting a kayak costs 17 dollars plus 9 dollars for each hour. Find the cost of 3 hours. → ***44 dollars*** *78 — (17 + 9)(3): the flat 17 was charged three times [8.NSO.1.7: only the hourly price is repeated]; 29 — 17 + 9 + 3, every number added [B1G-M 8.NSO.1.7 misconception: keyword hunting]*
21. **Board 3.**

22–23. Board 4. A square floor of side 8 feet is tiled at 3.50 dollars a square foot. Installation is a flat 60 dollars. Find the total cost. → ***284 dollars*** *4,064 — 64(3.50 + 60): the flat fee was put inside the area group [8.NSO.1.7: a flat fee is not a rate]; 88 — 8(3.50) + 60: the side used instead of the area*

24–25. Board 5. Evaluate ½mv² when m = 40 and v = 3. Which is the value? → ***A — 180*** *(B) v² was read as 2v [8.NSO.1.3: an exponent is not a factor]; (C) the one half was dropped [8.NSO.1.7: every factor in the formula is used]; (D) the exponent was applied to the whole product, (½ · 40 · 3)² [8.NSO.1.7: the exponent belongs to the v alone] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

26–27. Board 6. A square patio of side 12 feet is sealed at 1.25 dollars a square foot. Delivery of the sealer is a flat 38 dollars. Which expression gives the total cost? → ***D — $12^2(1.25) + 38$*** *(A) the side was used where the area is needed [8.NSO.1.7: read what the rate is per]; (B) the prices were added to the area instead of multiplied by it [B1G-M 8.NSO.1.7 misconception: keyword hunting]; (C) the flat delivery charge was put inside the area group, so it is charged on every square foot [8.NSO.1.7: a flat fee is not a rate] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. A square garden has 4 sides, each √150 feet long. Find its perimeter, to the nearest hundredth of a foot. → ***48.99 feet*** *48 — the side was rounded to 12 before multiplying [Unit 2: round once, at the end]; 48.99 with no unit — the answer is in feet*

30–31. Board 8. Evaluate (a² − b) ÷ c when a = 7, b = 4, c = 5. → ***9*** *48.2 — only the 4 was divided by 5 [8.NSO.1.7: the bar groups the whole numerator]; 5.8 — 49 ÷ 5 done first, then 4 subtracted*

32–33. Board 9. One pizza order costs 13 dollars plus 2 dollars for each of 5 toppings. Another costs 13 dollars for each of 2 pizzas, plus 5 dollars for delivery. Find both totals, and say why the key words could not have told you. → ***23 dollars and 31 dollars*** *23 and 31 with no explanation — the item asks why [same key words, different structures: the explanation is the evidence]; 23 twice — the second structure was not read*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Misconceptions to Watch

Board 1. −12 — the exponent was applied to the 2 and not to the whole substituted value [8.NSO.1.7: substitute in parentheses]

Board 2. anything built from 480 — the unneeded figure was used [the formula takes a mass and a speed]

Board 3. 78 — (17 + 9)(3): the flat 17 was charged three times [8.NSO.1.7: only the hourly price is repeated]

Board 4. 4,064 — 64(3.50 + 60): the flat fee was put inside the area group [8.NSO.1.7: a flat fee is not a rate]

Board 5. v^2 was read as 2v [8.NSO.1.3: an exponent is not a factor]

Board 6. the side was used where the area is needed [8.NSO.1.7: read what the rate is per]

Board 7. 48 — the side was rounded to 12 before multiplying [Unit 2: round once, at the end]

Board 8. 48.2 — only the 4 was divided by 5 [8.NSO.1.7: the bar groups the whole numerator]

Board 9. 23 and 31 with no explanation — the item asks why [same key words, different structures: the explanation is the evidence]

Materials Whiteboards and markers. Calculators are allowed on everything and are the right tool for the BMI and kinetic-energy items; the work being assessed is the expression, not the arithmetic.